

Learning Causal Variables, Not Causal Directions

Structured sparse autoencoders for nonlinear causal discovery

Validated on 5 GPT-2 tasks from the MIB benchmark

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Overview

Part 1: Setup

1. What is a causal variable?
2. DAS and the Grassmannian
3. Why linearity matters

Part 2: The atlas

4. 14 operations, three classes
5. Linear DAS fails completely
6. Stochastic grokking

Part 3: The vacuity problem

7. Memorization artifacts
8. Nonlinear DAS is vacuous
9. Intervention faithfulness

Part 4: The fix

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13. Subspaces to factors
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Part 5: Implications

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Part 1

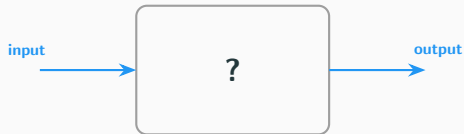
What is a causal variable, and where does DAS look?

1. What is a causal variable?

The interpretability goal

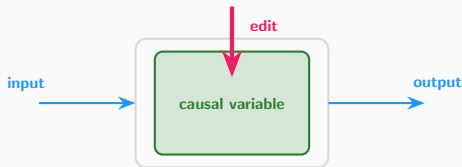
We want to understand *how* neural networks compute, not just *what* they compute.

Standard evaluation



Report accuracy, F1, etc.
Does not explain *why*

Causal abstraction



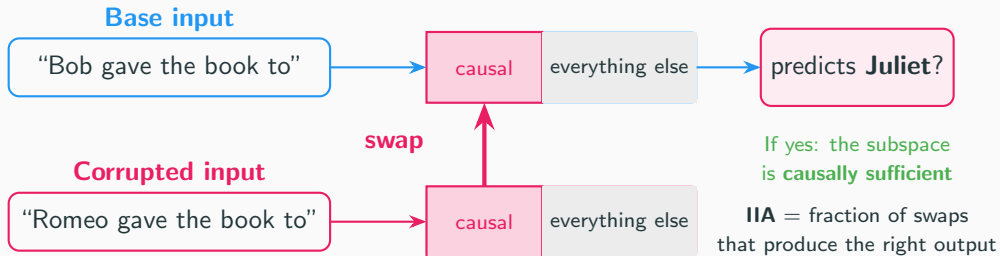
Find internal variables.
Edit them. Check if output changes.

If editing a specific direction in the network's internal representation changes the output predictably, that direction is a **causal variable** for the behavior.¹

¹Geiger et al., "Causal Abstraction for Faithful Model Interpretation," 2021. [arXiv:2301.04709](https://arxiv.org/abs/2301.04709)

Interchange intervention analysis (IIA)

The core technique: take two inputs, swap part of their internal representations, check if the output changes accordingly.²



²Geiger et al., "Inducing Causal Structure for Interpretable Neural Networks," ICML 2022. [arXiv:2112.00826](https://arxiv.org/abs/2112.00826)

2. DAS and the Grassmannian

What DAS actually does

DAS learns a rotation matrix $Q \in \mathbb{R}^{d \times k}$ that identifies a k -dimensional subspace of the d -dimensional activation space.^a

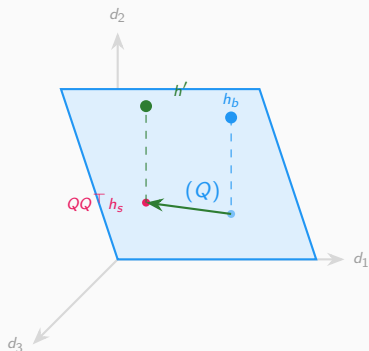
The interchange intervention projects out the base's component and replaces it with the source's:

$$h' = h_b - QQ^\top h_b + QQ^\top h_s$$

This is a **linear** operation. The subspace $QQ^\top$ is an orthogonal projection onto a *flat plane* in activation space.

The set of all k -dimensional planes in $\mathbb{R}^d$ is the **Grassmannian manifold** $\text{Gr}(k, d)$.

Every DAS solution is a point on $\text{Gr}(k, d)$.



DAS projects onto a **flat plane**.

What if the variable lives on a **curve**?

The Grassmannian manifold $\text{Gr}(k, d)$

The Grassmannian $\text{Gr}(k, d)$ is the set of all k -dimensional linear subspaces of $\mathbb{R}^d$.

Think of it as a “space of planes.” Every DAS solution picks one plane from this space. The distance between two planes is measured by their **principal angles**:

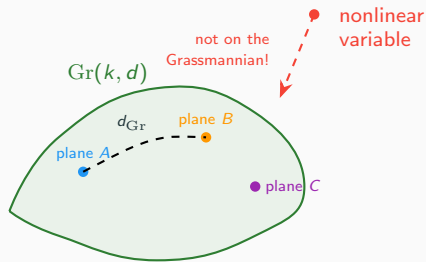
$$d_{\text{Gr}}([Q_1], [Q_2]) = (\sum_i \theta_i^2)^{1/2}$$

When we ask “does DAS work?” we are really asking:

Does the true causal variable live on $\text{Gr}(k, d)$?

If yes: DAS finds a meaningful point.

If no: DAS still returns a point, but it is an **artifact**.



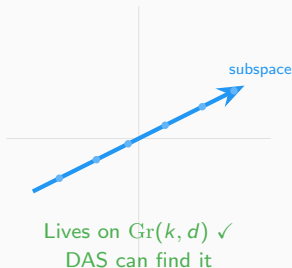
A nonlinear causal variable (circle, sphere)

does not live on $\text{Gr}(k, d)$ at any k .

3. Why linearity matters

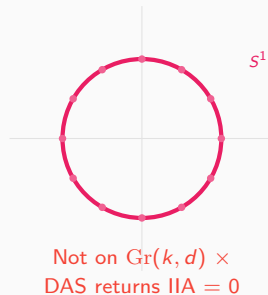
Linear vs. nonlinear causal variables

Linear causal variable



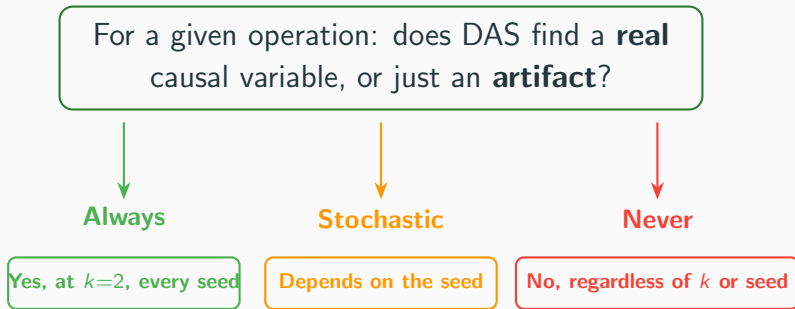
Examples: IOI indirect object (partially),
multiplication mod p after grokking

Nonlinear causal variable



Examples: addition mod p after grokking
(Fourier representation encodes numbers on a
circle)

The central question



We test fourteen modular arithmetic operations and find a **sharp, three-class partition**.

Why subspaces matter for circuits

Mechanistic interpretability asks two questions about neural network circuits:

Q1: Which components?

Which attention heads and MLPs matter for a behavior? Which edges carry the signal?

Methods: activation patching, attribution patching, EAP.

Q2: What do they communicate?

When two components are connected, *what* flows between them? Which subspaces of the residual stream carry the signal?

Methods: DAS, probing, SAEs.



Q1: *this edge matters*



Q2: *subspace s_i carries the signal*

Our work: whether Q2 has a **linear** answer depends on the operation.

Part 4

The fix: structured VAE and cross-task transfer

10. Structured VAE

Structured VAE for nonlinear causal variables

The structured VAE partitions the latent space into:

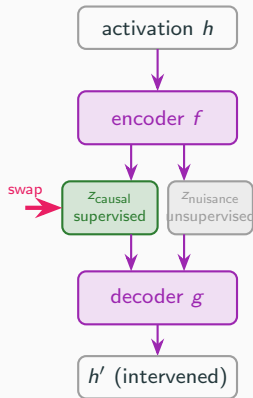
- z_{causal} : supervised with the operation's label
- z_{nuisance} : unsupervised, absorbs everything else

The loss combines three terms:

$$\mathcal{L} = \underbrace{-\mathbb{E}_q[\log p(x | y, z)]}_{\text{reconstruction}} + \beta \underbrace{D_{\text{KL}}}_{\text{regularization}} + \alpha \underbrace{\mathcal{L}_{\text{CE}}}_{\text{supervision}}$$

The ELBO prevents the lookup-table failure:

1. Reconstruction forces faithful decoding
2. KL prevents latent collapse
3. Causal/nuisance split preserves base info



Why the VAE is not vacuous: three constraints

NL-DAS (vacuous)

- × No reconstruction loss
- × No KL regularization
- × No causal/nuisance split
- × Trained only on IIA
- Can learn lookup table
- Diversity $\rho \approx 0$

VS.

Structured VAE (genuine)

- ✓ Reconstruction → faithful decoding
- ✓ KL → prevents latent collapse
- ✓ Causal/nuisance split
- ✓ ELBO + supervision
- Cannot learn lookup table
- Diversity $\rho \approx 1$

Controls confirm the VAE is not cheating

Random-label: $\text{IIA} \leq 0.02$ | Recon-only ($\alpha=0$): $\text{IIA} \leq 0.005$ | Untrained: $\text{IIA} = 0$
Plain VAE (no split): $\text{IIA} \leq 0.08$ — Supervision AND structure both required

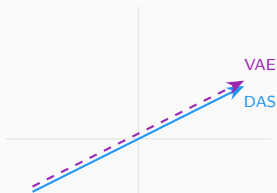
Khemakhem et al. 2020 (iVAE): identifiability guarantee

Label-conditional prior + injective decoder + ELBO $\Rightarrow$ recovers true causal variable up to component-wise bijection. NL-DAS violates injectivity $\rightarrow$ no guarantee.

VAE matches DAS where DAS works, extends where it fails

Always Grassmannian operations:

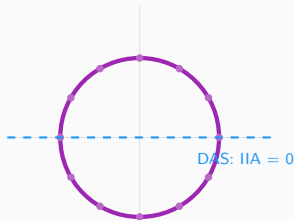
VAE reduces to an effective rotation.



Both find the same
linear subspace

Never Grassmannian operations:

VAE's nonlinear encoder captures curved structure.



VAE captures nonlinear
causal structure

For grokked addition and multiplication, both DAS and VAE achieve $\text{IIA} = 1.000$ and equivariance $> 99\%$.

The VAE is a **strict generalization** of DAS: when the variable is linear, $\text{VAE} = \text{DAS}$.

10b. Structured pi- π -SAE: sparse nonlinear causal discovery

From pi-VAE to Structured pi-SAE: sparsity + structure

pi-VAE (Zhou & Wei, 2020)

Label-conditional prior:

$$p(z \mid y) = \mathcal{N}(\mu(y), \sigma^2(y))$$

Each label gets its own mean and variance.
Forces the latent space to organize by label.

Published method. Works well on IOI.

But on grokked modular arithmetic:

IIA = 0 across all k .

Structured pi-SAE (ours)

pi-VAE + **causal/nuisance split** + **L1 sparsity**:

$$\mathcal{L} = \mathcal{L}_{\text{pi-VAE}} + \lambda \|z_c\|_1, \quad z = (z_c, z_n)$$

Explicit causal/nuisance partition forces the model to separate task-relevant from task-irrelevant variation.

L1 sparsity on z_c prevents memorization — the encoder can't just copy the input.

Two modifications over pi-VAE: (1) causal/nuisance split in the latent space, (2) L1 sparsity on causal dimensions. Neither alone suffices.

Structured pi-SAE vs. pi-VAE: results

IIA at $k=2$. Controls ≤ 0.02 .

	pi-VAE	Str. pi-SAE
Addition	0.000	1.000
Multiplication	0.000	1.000
Quartic sum	0.000	1.000
IOI (GPT-2)	0.945	0.980
Mixed product	0.010	0.040
Squaring	0.000	0.000

Non-grokked operations correctly return ≈ 0 .

What changed:

pi-VAE gets IIA = 0 on every grokking task.

The structured prior alone is not enough — the latent space collapses without the sparsity constraint.

Adding the causal/nuisance split + L1:

- All grokked operations $\rightarrow$ IIA = 1.0
- IOI $\rightarrow$ IIA = 0.98
- Non-grokked $\rightarrow$ IIA ≈ 0 (correct rejection)

Sparsity is the missing ingredient.

Intrinsic dimension: $k=1$ reveals the true structure

A circle is **1-dimensional** — one angle θ .

Linear DAS needs $k=2$: embeds as

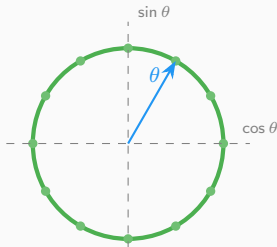
$(\cos \theta, \sin \theta)$. A nonlinear method learns $z = \theta$ directly.

Structured pi-SAE at $k=1$:

	DAS	NL-DAS	Str. pi-SAE
Addition	0.006	0.194	0.917
Multiplication	0.022	0.217	0.011
Quartic sum	0.011	0.244	0.106
IOI	0.185	1.000	0.988

Addition's circle admits a smooth 1D parameterization.

Multiplication's discrete-log circle does not.



Linear: needs $k=2$

Nonlinear: needs $k=1$

Linear DAS **overestimates**
intrinsic dimension.

Structured pi-SAE results across all operations

Operation	Grok	Acc	IIA at $k=2$			IIA at $k=1$		
			DAS	NL-DAS	Str. pi-SAE	DAS	NL-DAS	Str. pi-SAE
Addition	✓	1.00	0.06	0.49	1.00	0.01	0.19	0.92
Multiplication	✓	1.00	0.08	0.72	1.00	0.02	0.22	0.01
Quartic sum	✓	1.00	0.03	0.79	1.00	0.01	0.24	0.11
IOI (GPT-2)	—	—	0.30	1.00	0.98	0.19	1.00	0.99
Mixed product	×	0.15	0.02	0.60	0.04	0.01	0.21	0.02
Sym. power	×	0.21	0.02	0.63	0.02	0.01	0.26	0.02
Dbl. add mult	×	0.10	0.00	0.77	0.02	0.00	0.22	0.02
Squaring	×	0.01	0.00	0.01	0.00	0.00	0.01	0.00

Green bold: genuine causal variable found. Red: NL-DAS vacuity — high IIA despite no learned structure.

Str. pi-SAE: IIA = 1.0 on all grokked, ≈ 0 on all non-grokked. NL-DAS: 0.6–0.8 on non-grokked — **false positives**.

Ablation: both sparsity and structure are required

IIA at $k=2$: $\{\text{structured, plain prior}\} \times \{\text{VAE, SAE}\}$.

Operation	Plain prior $\mathcal{N}(0, I)$		Structured prior $p(z y)$	
	VAE	SAE	pi-VAE	Str. pi-SAE
Addition	0.02	1.00	0.00	1.00
Multiplication	0.00	0.72	0.00	1.00
Quartic sum	0.02	0.32	0.00	1.00
IOI	0.56	0.83	0.95	0.98
Mixed product	0.03	0.01	0.01	0.04
Squaring	0.00	0.00	0.00	0.00

Plain SAE works on addition but degrades on multiplication and quartic sum.

Structured pi-SAE is uniformly best. Prior + sparsity = robust across all tasks.

Neither component alone suffices: pi-VAE (no sparsity, no split) = 0; SAE (no structured prior) is inconsistent.

Structured pi-SAE exposes NL-DAS vacuity

The problem: NL-DAS achieves IIA
= 0.6–0.8 on operations that **never learned**
the task (acc < 20%). These are false
positives — the encoder learned a lookup
table.

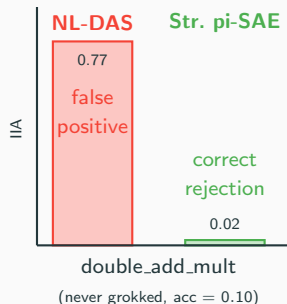
Structured pi-SAE correctly rejects: IIA
 ≤ 0.04 on the same operations.

Reconstruction prevents hallucination.

Reconstruction MSE confirms:

	Grokked	Not grokked
Recon MSE	0.6–1.2	11–24
Classifier acc	1.00	0.06–0.63

High MSE + low acc = no structure to recover.



IOI (GPT-2):

NL-DAS: IIA = 1.0, $\rho \approx 0$

CRT composite modulus: the first task needing $k > 1$

Chinese Remainder Theorem.

For composite modulus $n = p \times q$:

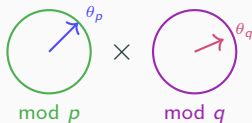
$$\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/q\mathbb{Z}$$

The model learns **two independent circles** — one mod p , one mod q .

Intrinsic dimension = **2** (two angles θ_p, θ_q).

Linear DAS needs $k \geq 4$ (cos / sin for each).

Structured pi-SAE captures both at $k=2$.



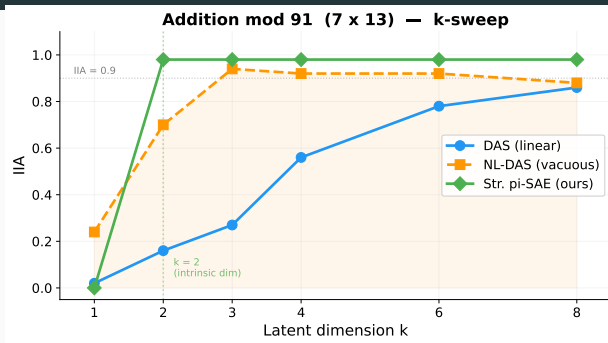
Task	k	DAS	NL-D	Str. pi-SAE	
add 113	1	0.01	0.19	0.92	✓
(prime)	2	0.06	0.49	1.00	
add mod 77	1	0.02	0.19	0.06	✓
(7 × 11)	2	0.07	0.59	0.99	
add mod 91	1	0.02	0.24	0.00	✓
(7 × 13)	2	0.16	0.70	0.98	
add mod 35	2	0.08	0.51	0.21	×
(5 × 7)	4	0.21	0.95	0.21	

Green: genuine. **Red:** NL-DAS false positive.

$k=1$ fails on CRT tasks $\Rightarrow$ intrinsic dim is truly 2.

Structured pi-SAE $k=2$ beats DAS $k=8$ (0.86–0.89).

CRT k -sweep: linear DAS needs $4\times$ more dimensions



Orange shading: NL-DAS false-positive region.

Structured pi-SAE saturates at $k=2$ (intrinsic dimension).

DAS climbs linearly — still hasn't converged at $k=8$.

Intrinsic dimension hierarchy: prime vs. composite

Min k for $\text{IIA} \geq 0.9$ reveals the **true intrinsic dimension**:

Operation	Structure	Min k	Manifold
add mod 113	prime	1	S^1
add mod 77	7×11	2	$S^1 \times S^1$
add mod 91	7×13	2	$S^1 \times S^1$
IOI (GPT-2)	NLP	1	$S^1?$

Prediction: mod $p_1 \cdots p_n$ needs $k = n$.

Testable: mod $2310 = 2 \times 3 \times 5 \times 7 \times 11 \rightarrow k=5$.

Linear DAS always overestimates:

	Linear k	Nonlinear k
Prime (S^1)	2	1
Composite ($S^1 \times S^1$)	4+	2

$k = 1$: **one circle**



mod p (prime)

$k = 2$: **torus** $S^1 \times S^1$



Nonlinear methods reveal the manifold's true topology, not its linear embedding dimension.

10c. Beyond grokking: NLP tasks on GPT-2

Structured pi-SAE on GPT-2 language tasks

Apply to **5 NLP tasks** in GPT-2-small, layer 10.

Using MIB counterfactual pairs (Hanna et al. 2024):

- **SVA**: singular/plural agreement (12,400 pairs)
- **Greater-than**: year comparison (1,000 pairs)
- **Gender bias**: occupation → he/she (986 pairs)
- **Capitals**: country → capital (190 pairs)
- **Hypernymy**: word → category (198 pairs)

Does pi-SAE's conservatism hold on language tasks?

Task	DAS	NL-DAS	Str. pi-SAE
$k = 1$			
SVA	0.00	1.00	0.00
Greater-th	0.03	1.00	0.00
Gender bias	0.88	1.00	0.54
Capitals	0.00	0.00	0.12
Hypernymy	0.03	0.20	0.50
$k = 4$			
SVA	0.02	1.00	0.00
Greater-th	0.31	1.00	0.00
Gender bias	1.00	1.00	0.95
Capitals	0.02	0.07	0.37
Hypernymy	0.08	0.43	0.47

Red: NL-DAS false positives.

Three regimes of causal structure

1. High-dimensional

SVA, greater-than

All methods except NL-DAS get ≈ 0 .

NL-DAS gets 1.0 = **false positive**.

Pi-SAE correctly returns 0:

no low-dim structure exists.

2. Genuinely low-dim

Gender bias

Binary task (he/she), linearly accessible.

DAS: 0.88–1.0.

Pi-SAE: 0.54–0.95.

Both find it, DAS is slightly better on this linear task.

3. Nonlinear structure

Capitals, hypernymy

DAS fails (≈ 0).

NL-DAS also weak.

Pi-SAE is the only method that finds structure.

Capitals: 0.37 at $k=4$.

Hypernymy: 0.50 at $k=1$.

Pi-SAE is **conservative**: returns 0 when no structure exists, finds structure where linear methods can't.

Factorized DAS matches optimal DAS exactly

Hard-mode IIA, layer 8, $k=1$,
atomic-sweep-40:

Task	DAS	Fac. DAS
IOI	0.94	0.94
SVA	0.79	0.79
Gender	0.79	0.83
Capitals	0.69	0.69
Hypernymy	1.00	1.00

Factorized DAS recovers the same subspace as
optimal DAS on every task.

Complementary tools:

	DAS	Str. pi-SAE
Linear tasks	✓	✓
Nonlinear	×	✓
False positives	low	none

DAS identifies subspaces (linearly accessible variables).

Pi-SAE detects structure (including nonlinear).

Together: structure detection + subspace identification.

What do the features learn?

Most features are causally inert

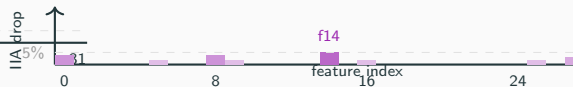
Addition (grokked, 32 features)



3 of 32 features account for 97% of causal effect.

26 features have **zero** IIA impact when ablated.

Gender bias (GPT-2, 32 features)



Feature 28 alone causes 0.21 IIA drop.

Binary task → one dominant feature (the he/she direction).

Sparse features are **genuinely sparse**. Ablation reveals which carry causal information and which are inert.

Grokking features align with Fourier modes

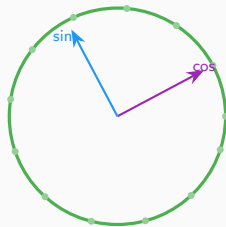
For addition mod 113, all 32 features align to **frequency 9**:

	ρ	Freq.	Component
f16 (rank 1)	0.55	9	cos
f19	0.58	9	cos
f1 (rank 3)	0.48	9	sin
f15	0.48	9	sin
f11	0.43	9	sin
f6	0.39	9	sin

31 of 32 features: frequency 9

Confirms Nanda et al. (2023): grokked addition uses Fourier components of $s = (a+b) \bmod 113$.

The pi-SAE **rediscovers** this without being told about Fourier modes.



S^1 : **freq. 9 of $(a+b) \bmod 113$**
Features encode cos and sin
of the 9th Fourier mode

Multiplication: Fourier $\rho < 0.03$.

Multiplicative structure $\neq$ clean Fourier modes.

What do the features detect? Max-activating examples

Gender bias: feature 28 (dominant, IIA drop: 0.21)

Act.	Text
+3.59	The assassin wanted that
+3.49	The collector was promoted because
+3.44	The tutor yelled that
-3.43	The actress was promoted because
-3.43	The performer yelled that
-3.43	The teenager ate because

Positive: male-stereotyped occupations.

Negative: female-stereotyped occupations.

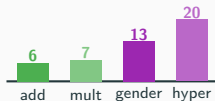
Feature 28 is the gender direction.

Hypernymy: distributed features

Feature	IIA drop	Role
f8	0.167	primary
f4	0.100	secondary
f10	0.083	secondary
f1	0.050	tertiary
f6	0.050	tertiary

20 of 32 features contribute

Active features per task



Grokking: **few features, high sparsity**. NLP: **more features, distributed**. The pi-SAE adapts to each task's intrinsic complexity.

End-to-end intervention training: the breakthrough

Key insight: train Structured pi-SAE with **differentiable intervention CE loss** — backprop through the model's own logits after intervention.

Two-phase training:

1. Warmup (200 epochs): reconstruction + classification only
2. E2E (300 epochs): add
— $\log p(\text{src_token} \mid \text{intervened})$

This is what DAS does implicitly — train on the intervention objective. But we combine it with **structured sparsity** and **nonlinear** encoding.

Hypernymy — standard IIA (strict IIA)		
Method	$k = 1$	$k = 4$
DAS	0.07 (0.02)	0.23 (0.08)
NL-DAS	0.72 (0.15)	0.92 (0.32)
Vanilla VAE	0.00 (0.00)	0.48 (0.18)
pi-VAE	0.00 (0.00)	0.57 (0.20)
Str. pi-SAE	0.58 (0.17)	0.52 (0.13)
Str. pi-SAE E2E	0.97 (0.90)	0.98 (0.85)
Gender bias — standard IIA (strict IIA)		
Method	$k = 1$	$k = 4$
DAS	0.62 (0.51)	0.88 (0.85)
NL-DAS	1.00 (1.00)	1.00 (1.00)
Str. pi-SAE	0.41 (0.35)	0.52 (0.46)
<i>Str. pi-SAE E2E</i>	<i>pending</i>	<i>pending</i>

Why E2E works: closing the training-evaluation gap

Without E2E (classification loss only):

- Encoder learns to classify labels
- Decoder learns to reconstruct activations
- But: **swapping** z_c and decoding may produce activations the model doesn't interpret correctly
- Reconstruction error $\neq$ intervention error

With E2E (intervention CE loss):

- Encoder/decoder jointly optimized so that **swapped codes produce correct model outputs**
- The model itself is the judge

Additive vs. replacement intervention:

Hypernymy $k=1$	Add.	Repl.
Str. pi-SAE	0.58	0.60
Str. pi-SAE E2E	0.97	0.97
Strict IIA		
Str. pi-SAE	0.17	0.18
Str. pi-SAE E2E	0.90	0.88

E2E training eliminates the gap between additive and replacement — the decoder learns to produce activations the model actually uses.

Key: E2E + structured sparsity is the combination. E2E alone (on vanilla VAE) would overfit; sparsity constrains the causal

11. Cross-task transfer

Cross-task transfer validates causal variables

A genuine causal variable should **transfer** — train on one task variant, test on another.

IOI cross-template transfer (GPT-2):

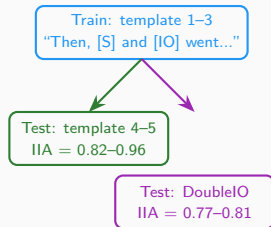
- Train VAE on templates 1–3
- Test on unseen templates 4–5
- IIA = 0.82–0.96 at $k = 4$

Extended IOI tasks^a:

- DoubleIO: IIA = 0.77–0.81
- TripleIO: IIA = 0.77–0.81
- Trained only on base IOI!

^aNainani et al., arXiv:2411.16105, 2024.

Transfer validation



A lookup table cannot transfer like this

IOI subtask transfer^a (8 types):

DAS ratio: 0.56 · VAE ratio: 0.43

^aPrakash et al., ICLR 2024 (MIB).

Cross-task transfer: train on A, evaluate on B

Train Structured pi-SAE on one task, freeze encoder/decoder, evaluate IIA on a different task's counterfactual pairs.

IIA transfer matrix ($k=4$, layer 8, additive):

Train \ Eval	Gender	Hyper.	Gt.
Gender bias	1.00	0.00	0.41
Hypernymy	0.30	0.98	0.43
Greater-than	0.31	0.00	1.00

Findings:

- Gender $\leftrightarrow$ greater-than: partial transfer (~ 0.3 – 0.4)
- Hypernymy is isolated — 0.0 transfer with gender
- Diagonal dominates: task-specific structure is real

Subspace geometry:

Task pair	$\cos \theta$	d_G
Gender–Hyper.	0.11	2.92
Gender–Gt.	0.09	2.96
Hyper.–Gt.	0.09	2.96

Grassmannian distance $d_G \approx 2.9$. The decoder causal subspaces are **nearly orthogonal**.

Interpretation: each task occupies its own region of the residual stream. Partial transfer (gender $\leftrightarrow$ greater-than) comes from **shared numerical structure**, not shared subspaces.

Consistent with superposition — GPT-2 packs many independent causal variables into the same residual stream.

Feature composition: DAS vs. pi-SAE

DAS: direction arithmetic

Linear DAS learns a rotation Q per task. To compose:

- $h' = h + Q_A^\top z_A + Q_B^\top z_B$ (add both)
- $h' = h - Q_A^\top z_A$ (negate A)

Problems:

- Directions may not be orthogonal
- Adding two continuous rotations can interfere
- Only works if structure is linear

Structured pi-SAE: feature switching

Each sparse feature is an independent binary knob:

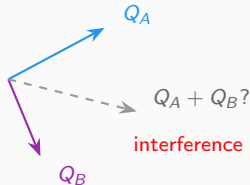
- Activate features 3 and 7: both effects
- Zero feature 5: negate that effect
- Set feature 3 to specific value: titrated control

Sparsity guarantees near-independence by construction.

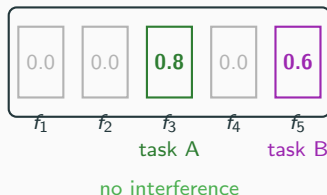
Sparse features compose without interference. Continuous directions do not.

Sparse features as independent causal knobs

DAS: add directions



pi-SAE: set features



Next: train pi-SAE on tasks A and B, compose by setting features from both, measure joint IIA.

If causal features are truly independent, composition should “just work.”

13. Four-level hierarchy

A four-level hierarchy for causal validity

Level 1: Interchange effectiveness (IIA)

Necessary, not sufficient. Squaring gets $\text{IIA} = 1.0$ without structure.

Level 2: Interchange faithfulness (ρ , recon)

Kills NL-DAS. Automatic for linear DAS. Required for nonlinear.

Level 3: Distributional fidelity (KL, JS, logit diff)

Distinguishes quality among methods with $\text{IIA} \approx 1.0$.

Level 4: Structural diagnostics (equiv, transfer)

Strongest evidence. Equivariance or cross-distribution generalization.

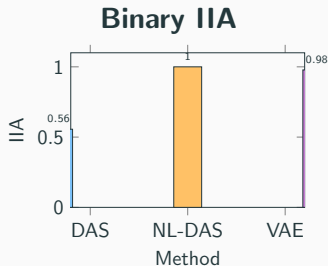


stronger evidence

Key asymmetry: Linear DAS automatically passes Level 2 (orthogonal projection is faithful by construction) but may fail Level 1 on nonlinear variables. NL-DAS easily passes Level 1 but fails Level 2.

14. Continuous metrics

Binary IIA hides important differences



NL-DAS “wins.” But it is a lookup table.

With faithfulness metrics

	DAS	NL-DAS	VAE
IIA	0.556	1.000	0.978
Diversity ρ	1.0	≈ 0	≈ 1
Transfer	Yes	No	Yes

VAE wins genuinely. NL-DAS is disqualified
by $\rho \approx 0$.

DAS is honest but limited by linearity.

15. Practical advice

Our flowchart for causal abstraction

1. **Never report IIA alone.** $\text{IIA} = 1.0$ is compatible with genuine structure, memorized lookup tables, and degenerate encoder-decoders. Always report at least one faithfulness metric.
2. **Use continuous distributional metrics.** KL divergence and normalized logit difference distinguish among methods that all achieve $\text{IIA} \approx 1.0$. Binary IIA treats a 51% shift the same as 99%.
3. **Be suspicious of unconstrained nonlinear featurizers.** Any encoder-decoder trained end-to-end on interchange loss without reconstruction constraints can learn a lookup table. The ELBO (or equivalent) is necessary.
4. **Test cross-distribution generalization.** Evaluate on disjoint names, novel templates, or new task variants. A method that fails under distribution shift is exploiting surface regularities.
5. **Consider nonlinear methods when DAS IIA is low.** Low IIA does not mean no causal variable — it may mean DAS is searching the wrong function class. But validate with faithfulness metrics.

Surprising positive and negative cases

Surprising positives

Cubic sum ($a^3 + b^3$): perfect equivariance (100%) despite cubing alone reaching only 50.9%. The boundary is not “group action vs. polynomial” but whether the binary structure provides sufficient additive symmetry.

Max ($\max(a, b)$): achieves 95.7% equivariance despite not being a group action. It is piecewise equivariant: when the argmax is stable under shift, max behaves like a shifted identity.

Surprising negatives

Affine ($2a + 3b + 5$): a *linear* function that *fails* to produce a Grassmannian variable! Under $a \mapsto a + g$, the output shifts by $2g$, not g . Equivariance requires the operation to be a group action, not merely linear.

Squaring: achieves $\text{IIA} = 1.0$ by $k = 4$ without grokking. High IIA is the illusion; low equivariance (31.4%) is the truth.

16. Conclusions

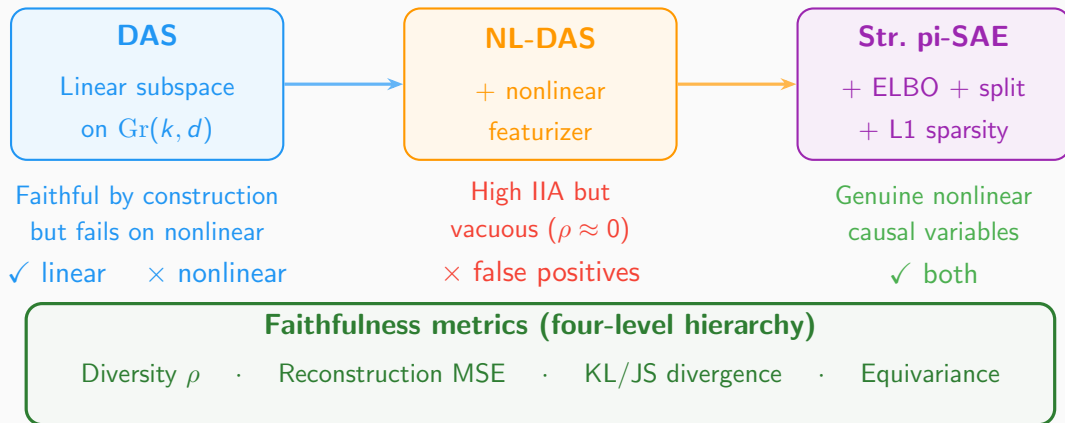
Conclusions

1. 14 operations $\rightarrow$ three sharp classes (Always / Stochastic / Never Grassmannian).
Boundary governed by grokking.
2. Linear DAS returns $\text{IIA} = 0$ on grokked addition. The variable lives on S^1 , not $\text{Gr}(k, d)$.
Category error.
3. Unconstrained NL-DAS achieves $\text{IIA} = 1.0$ vacuously (lookup table). Diversity ratio ρ exposes this.
4. Structured VAE recovers genuine nonlinear causal variables. ELBO prevents degeneracy.
Transfers across tasks.

The linearity assumption is **testable**. The natural fix (nonlinearity) introduces a new failure mode. This paper provides the diagnostics to detect both problems and the method to avoid them.

Summary

Does model M represent causal variable Z ?



IIA is necessary but never sufficient. Every causal abstraction claim needs the **four-level hierarchy**.

Thank you

Elliot Tower

Code and experiments:

`github.com/elliottower/grokking-causal-geometry`

Draft paper: *When Does Linear Causal Abstraction Work?*

Mapping the Boundary on the Grassmannian